MA378 Chapter 3: Numerical Integration

§3.3 Precision and Composition

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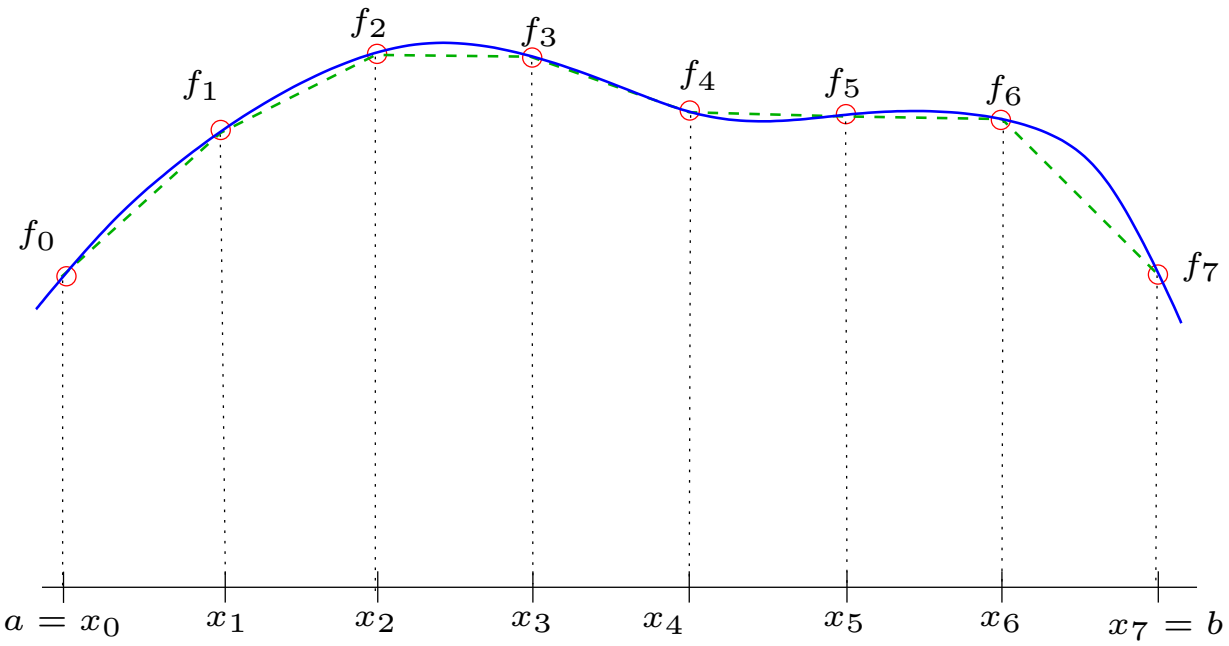


3.0 What, precisely, we'll discuss

1 Simpson's, again

- Details
- Errors

2 Precision

3 Composite Rules

4 Exercises

3.1 Simpson's, again

In the final example of Section 3.2 ("Simpson's Rule") we noticed that $\boxed{\text{Simpson's Rule}}$ yields exactly the right answer (i.e., the error is zero) when applied to approximating $\int_0^1 x^3 dx$.

In fact, Simpson's Rule is exact for **any** polynomial of degree 3 or less. We'll now explain why, restricting our focus to the interval $[-1, 1]$. (See below for the general case).

1. Suppose that $Q_2(f)$ approximates $\int_{-1}^1 f(x)dx$.
2. We know that, if p_2 is any quadratic polynomial, then $Q_2(p_2) = \int_{-1}^1 p_2(x)dx$.
3. Write p_3 as

$$p_3(x) = c_3x^3 + \underbrace{c_2x^2 + c_1x + c_0}_{p_2} = c_3x^3 + p_2(x),$$

for some quadratic $p_2(x)$.

3.1 Simpson's, again

4. Note that $\displaystyle\int_{-1}^{1} p_3(x)dx = c_3 \int_{-1}^{1} x^3 dx + \int_{-1}^{1} p_2(x)dx.$
5. However, $\int_{-1}^{1} x^3 dx = 0.$ ✓
6. Therefore, $\displaystyle\int_{-1}^{1} p_3(x)dx = \int_{-1}^{1} p_2(x)dx.$
7. Similarly, $Q_2(p_3)dx = c_3 Q_2(x^3) + Q_2(p_2).$
8. However, $Q_2(x^3) = \frac{2}{6}\big((-1)^3 + 4(0^3) + 1^3\big) = 0.$
9. Therefore $Q_2(p_3)dx = Q_2(p_2) = \displaystyle\int_{-1}^{1} p_2(x)dx.$ Since Q_2 is exact for all quadratics, by design.
10. So, we have that

$$\int_{-1}^{1} p_3(x)dx = \int_{-1}^{1} p_2(x)dx = Q_2(p_3).$$

Conclude: $Q_2(\cdot)$ gives exactly the right answer when applied to estimating $\int_{-1}^{1} p_3(x)dx$ for any cubic polynomial, p_3.

3.1 Simpson's, again

Details

Here are the details for the general case. We won't go though this in class, but please read it carefully.

We now claim that $\boxed{\text{Simpson's Rule}}$ is exact for *any* polynomial of degree 3 or less, and on *any* interval.

Denote by $\boxed{Q_2(f)}$ the approximation of $\int_a^b f(x)dx$ with Simpson's Rule:

$$Q_2(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

Since the method can be derived by integrating the quadratic that interpolates $f(x)$ at the three points a, $(a+b)/2$, and b, it is clearly exact for all quadratics.

3.1 Simpson's, again

Details

Let $f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$. Then

$$\int_a^b f(x)dx = \int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx + c_3 \int_a^b x^3 dx =$$
$$\int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx + c_3 \frac{b^4 - a^4}{4}.$$

Also,

$$Q_2(f) = Q_2(c_0 + c_1 x + c_2 x^2) + Q_2(c_3 x^3) =$$
$$\int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx$$
$$+ c_3 \left(\frac{b-a}{6}\right)\left(a^3 + 4\left(\frac{a+b}{2}\right)^3 + b^3\right). \quad (1)$$

3.1 Simpson's, again

Details

With a bit of symbolic manipulation we get that

$$\frac{b^4 - a^4}{4} = (\frac{b-a}{6})\left(a^3 + 4(\frac{a+b}{2})^3 + b^3\right),$$

as required.

In the previous section (§3.2 Simpson's Rule), we discussed a naive attempt to derive an upper bound for the error in Simpson's rule leading to

$$\mathcal{E}_2 \leq \frac{(b-a)^4}{196} M_3.$$

This is not wrong – just not sharp. For example it does not give that Simpson's Rule is exact for all cubic polynomials. The sharp result is

Theorem 3.1

$$\mathcal{E}_2 := \left| \int_a^b f(x)dx - Q_2(f) \right| \leq \frac{(b-a)^5}{2880} M_4.$$

For the proof see the text book (Theorem 7.2 of Suli and Mayers). Instead of working through it in class we'll prove a more general version of a consequence it.

3.2 Precision

Definition 3.2 (Precision of a Quadrature Rule)

A quadrature rule has **precision** N if it is exact for all polynomials of degree N or less. That is, the rule $Q(f)$ has precision N if

$$Q(p) = \int_a^b p(x)dx \quad \text{for all } p \in \mathcal{P}_N.$$

Example 3.3

By construction, the $(N+1)$-point Newton-Cotes rule has precision N.

thanks to the uniqueness of polynomial interpolation.

Eg: Trap Rule has Precison 1. By design
Simpson's Rule has precision 2, but in fact achieves precision 3.

 Note: this is just Q_N with N even.

Theorem 3.4

If $Q_{2k}(\cdot)$ is a Newton-Cotes quadrature rule on $2k+1$ points, then $Q_{2k}(\cdot)$ has in fact precision $2k+1$.

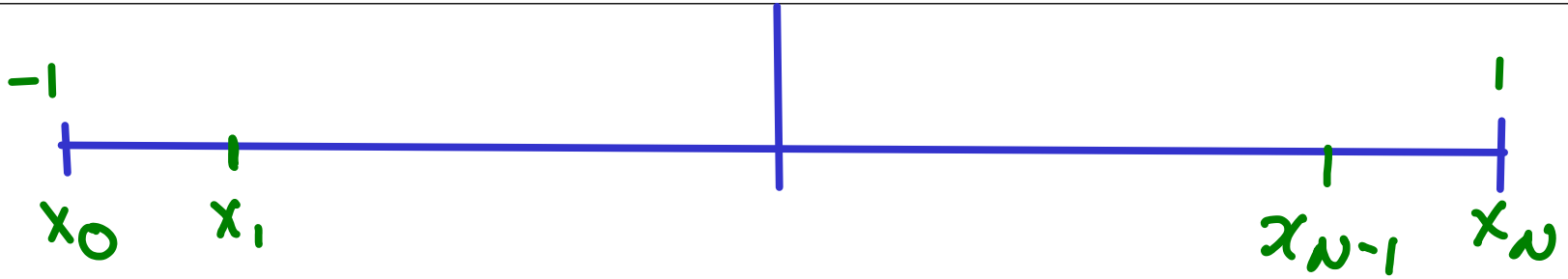
Proof: Let p be a polynomial of degree $N+1$ (where $N=2K$). We wish to show that $Q_N(p) = \int_a^b p(x)dx$. We can take $a=-1$, $b=1$ because a simple linear transformation can be used to map to an arbitrary interval.

Also, since the quadrature points are equally spaced on $[-1,1]$ we have that $\cancel{x_i = -x_{n-i}}$. $x_i = -x_{N-i}$

Furthermore (see Exercise 3.1) the quadrature weights are symmetric: $\cancel{q_i = q_{n-i}}$. $q_i = q_{N-i}$

Tip: $q_i = \int_{-1}^{1} L_i(x)\,dx$

3.2 Precision



$x_0 = -x_N$

$x_1 = x_0 + h = -x_N + h = -x_{N-1}.$

And, in general $x_i = -x_{N-i}$

Let $p(x)$ be a polynomial of degree $N+1$, and we write

$$p(x) = c_{N+1} x^{N+1} + p_N(x)$$ where

$p_N(x)$ is some poly of degree N.

We know that

$$Q_N(P_N) = \int_{-1}^{1} P_N(x)\,dx$$

And $\int_{-1}^{1} x^{N+1}\,dx = 0$ since N is even.

So it remains to show

$Q_N(x^{N+1}) = 0$. To see this use

$$Q_N(x^{N+1}) =$$

$$q_0 x_0^{N+1} + q_1 x_1^{N+1} + \dots + \boxed{q_{N-1}} x_{N-1}^{N+1} + q_N x_N^{N+1}$$

$$q_0 x_0^{N+1} + q_1 x^{N+1} + \dots + q_1(-x_1)^{N+1} + q_0(-x_0)^{N+1} = 0$$

3.3 Composite Rules

Suppose that we want to estimate $\int_a^b f(x)dx$ and the Trapezium rule is not sufficiently accurate. We could try Simpson's Rule, which should be better. Failing that, we could try a 4-point rule, based on integrating the p_4 interpolant, or a five-point rule, based on integrating p_5.

However, quite apart from the fact that it might be tedious to derive these rules, we know (Runge's example again!) that high-order polynomial interpolation can be very inaccurate.

3.3 Composite Rules

It is better to use a **Composite Rule**. This is analogous to the idea behind piecewise linear interpolation.

For the **Composite Trapezium Rule** we divide $[a, b]$ into N intervals of size $h = (b - a)/N$. Applying the Trapezium Rule on each interval $[x_{i-1}, x_i]$ we get

$$\int_{x_{i-1}}^{x_i} f(x)dx \approx h\frac{f_{i-1} + f_i}{2}.$$

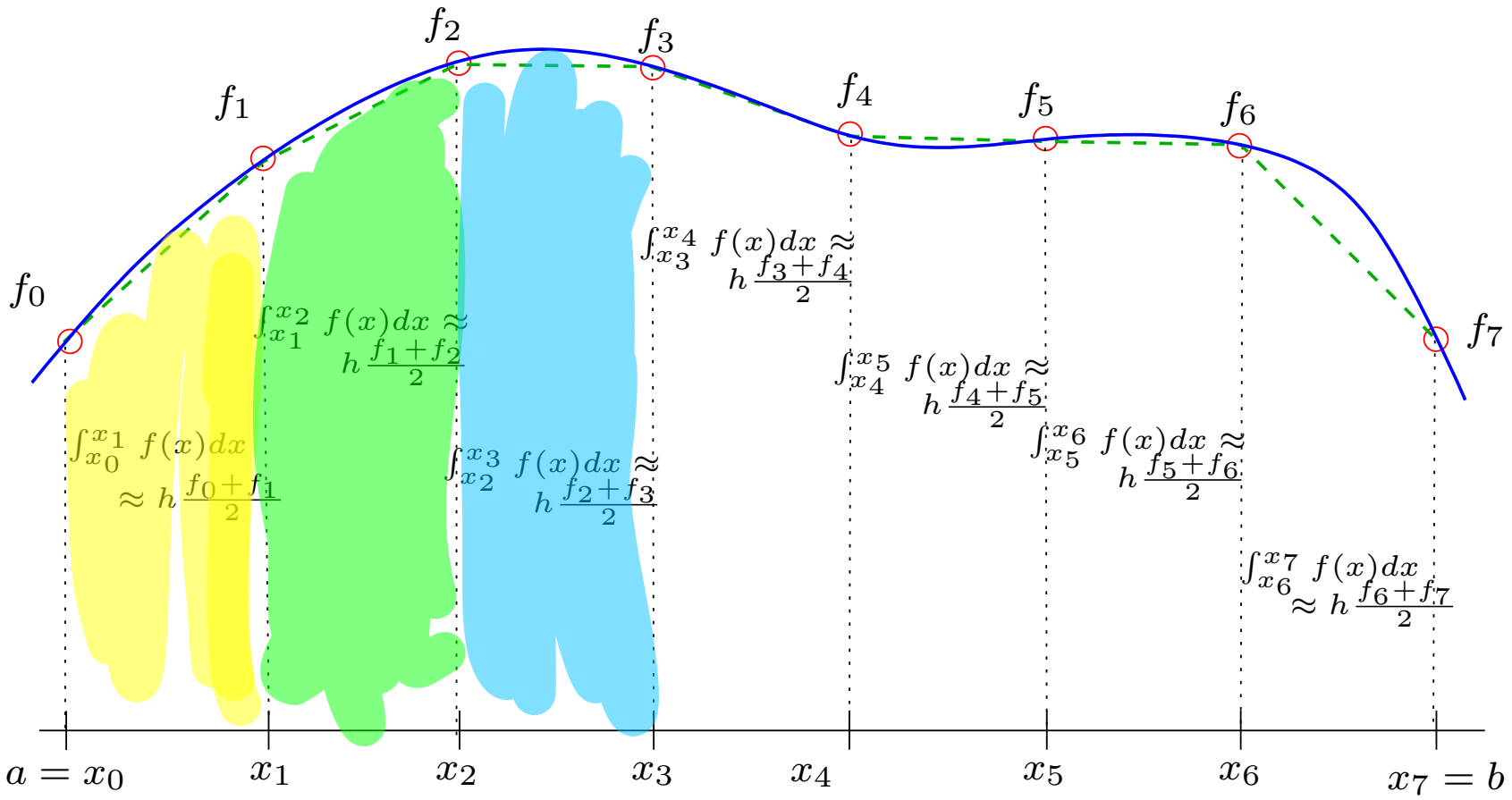
Summing for the N intervals we get

$$\int_a^b f(x)dx \approx \frac{b-a}{N}\left(\frac{f_0}{2} + f_1 + f_2 + \cdots + f_{N-1} + \frac{f_N}{2}\right). \qquad (2)$$

Note

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

3.3 Composite Rules



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3.4 Exercises

Exercise 3.1

Explain clearly, with an example, why in general it is not true that $Q_n(f) \to \int_a^b f(x)dx$ as $n \to \infty$.

Exercise 3.2

(i) Deduce an error estimate for the Composite Trapezium Rule.

(ii) Taking $N = 10$, give an upper bound for the error in the Composite Trapezium Rule when approximating $\int_1^2 \ln(x)dx$.

(iii) What value of N would you have to take to ensure that the error was less that 10^{-5}?

3.4 Exercises

Exercise 3.3

(i) Deduce the formula for the *composite Simpson's Rule*.

(ii) Derive an error estimate for the *composite Simpson's Rule*.

(iii) What value of N would you have to take to ensure that the error in the estimate of $\int_1^2 \ln(x)dx$ is less that 10^{-6}?

(iv) Denote the $(N+1)$-point Composite Simpson's Rule by $S_N(f) \approx \int_a^b f(x)dx$. Show that, for sufficiently smooth $f(x)$,

$$\lim_{n\to\infty} S_N(f) = \int_a^b f(x)dx.$$

3.4 Exercises

Exercise 3.4

Determine the precision of the following schemes for estimating $\int_0^1 f(x)dx$.

(i) $Q(f) = f(\frac{1}{2})$.

(ii) $Q(f) = \frac{1}{4}f(0) + \frac{3}{4}f(\frac{2}{3})$.

(iii) $Q(f) = \frac{3}{2}f(\frac{1}{3}) - 2f(\frac{1}{2}) + \frac{3}{2}f(\frac{2}{\mathbf{3}})$.

Exercise 3.5

Consider the rule:

$$R(f) = q_0 f(\frac{1}{3}) - f(\frac{1}{2}) + q_2 f(\frac{3}{4})$$

for approximating $\int_0^1 f(x)dx$.

(a) Determine values of q_0 and q_2 that ensure this rule has precision 2.

(b) What is the maximum precision of $R(\cdot)$ with the values of q_1 and q_2 that you have determined?

(c) Why is this not, strictly speaking, a Newton-Cotes rule?